CSC 436 Assignment

Semester DB Project: Phase II

Overview

Your group will analyze suggestions and critiques, document specific revisions, address any identified weaknesses, and implement revisions to your database structure and queries. Through this process, you will enhance the quality and functionality of your database.

Part I: Review Feedback & Revisions

Carefully review the feedback provided during the Phase II presentations and peer reviews. Take note of any suggestions, critiques, or areas for improvement identified in your database.

1. Thoroughly document the specific revisions and improvements to be made to your group's database. Include explanations for each change, detailing why it was made and how it enhances the database.

    a. Allow chemical quantity to be able to be 0, but can't be negative. It is made so that it allows inventory of chemical quantities to be 0, indicating empty inventory for such items. So that we do not have to remove the row each time it becomes 0. This will help the database to be more efficient.

    b. Hash sensitive chemical information. This change was made because some of the chemical information could be sensitive information, hashing such chemical information can help prevent unauthorized users from accessing these information from cybersecurity attacks.

    c. Remove Redundancy and clean up duplicate constraints. Some records in the Document table were duplicated and several tables had duplicate foreign key constraints referencing the same columns, including Batch, Document, Experiment, Inventory_item, and Usage_log. The revision is made to remove redundancy on the database and clean up duplicate constraints. It enhances the database by reducing the overall complexity of the database, increasing the maintainability of the database.

2. Identify weaknesses or concerns raised in the feedback and explain how your group will address them. Prioritize revisions that will have the most significant impact on improving the overall quality and functionality of your database.

Work closely with your group members to discuss the feedback received and determine the best approach for making revisions.

- One weakness was that our inventory rule did not reflect real lab workflow. A chemical can run out, but that does not mean it should disappear from the database. We fixed this by changing the quantity validation to allow 0.

- Another weakness was schema redundancy. Even though the peer review said our database was well structured and showed strong normalization overall, duplicated rows and repeated foreign key constraints still created unnecessary complexity. We addressed this by cleaning the schema and removing repeated definitions.

- A third weakness was consistency in user-entered text fields. If fields such as role or unit are not standardized, the system can slowly become messy and harder to query. We addressed this by enforcing more consistent allowed values.

Part II: Implementation Of Revisions

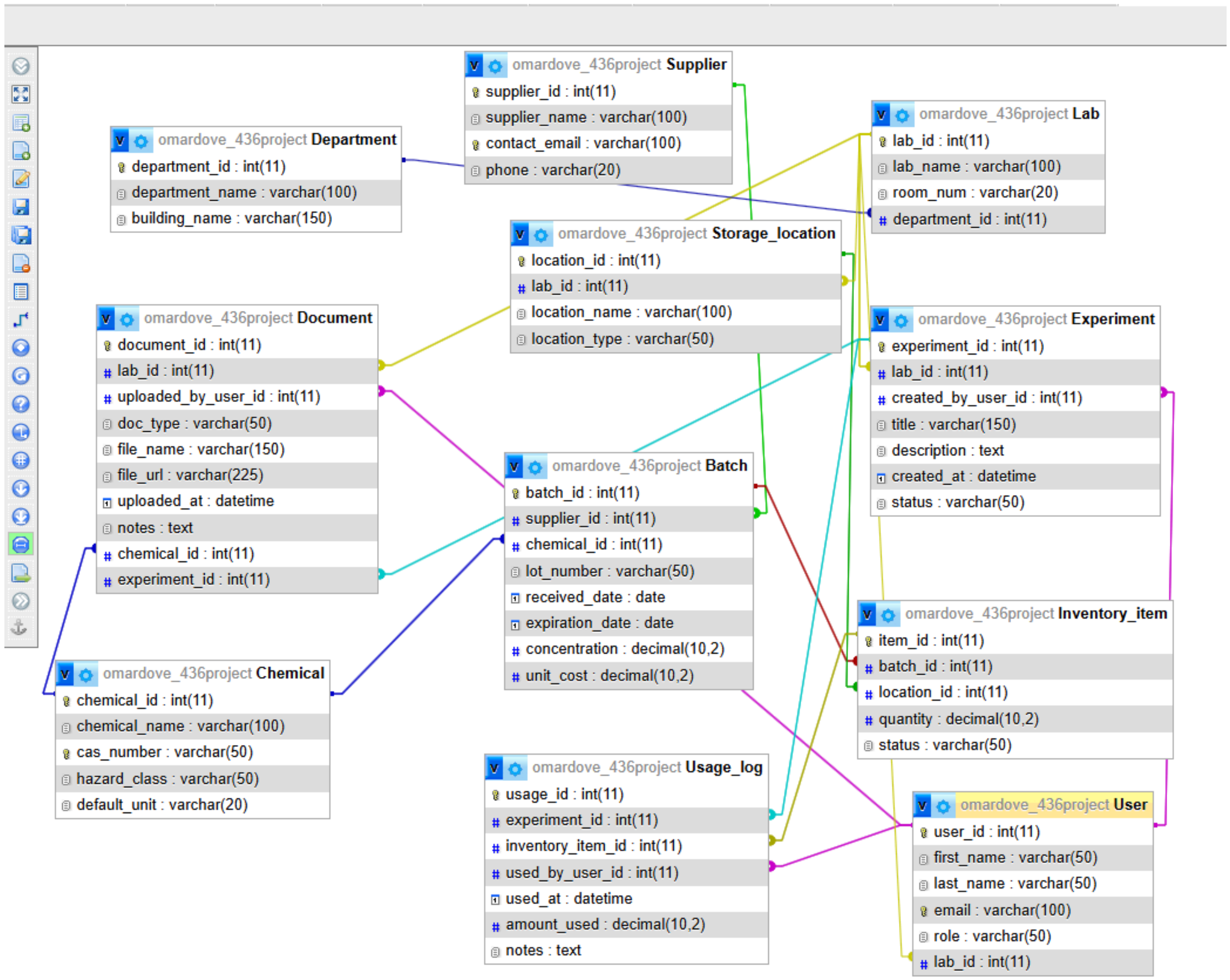
Implement the identified revisions and improvements into your database. Ensure that all changes are accurately reflected in your project materials. Your group must submit the following revised project materials based on the feedback you received:

3. Revise your database structure based on the feedback received and include explanations for the changes made. This may include updates to tables, integrity constraints, validation checks, and any proposed security measures. Provide screenshots of any modifications using the **Browse** tab to display updated data.

- The integrity constraint that kept quantity > 0 got changed to quantity >=0 which will allow chemicals to live inside of the database even if equal to 0. This is shown below on row 3.

item_id	batch_id	location_id	quantity	unit	status
1	1	1	4500.00	mL	In Stock
2	2	1	2000.00	mL	In Stock
3	3	2	0.00	g	In Stock
4	4	2	1500.00	mL	In Stock
5	5	2	3000.00	g	In Stock
6	6	3	2500.00	g	In Stock

- We removed redundant unit columns from Inventory_item and Usage_log, and made them still retrievable through JOIN instead of duplication.



4. Revise your SQL queries based on the feedback received and include explanations for the changes made. Provide the actual SQL queries written, screenshots of the tables displaying the query results, and insights extracted from these results to showcase the effectiveness of the revisions.

Alter Table - Updates the database structure by adding a new field to the Chemical table, and it lets the system store rules labeled such as “Keep Refrigerated" or “Store in flammable container”. This query executes “storage_requirment” into the Chemical table.

- ALTER TABLE Chemical
ADD COLUMN storage_requirement VARCHAR(100) ;
ALTER TABLE Chemical DROP COLUMN storage_requirement

	chemical_id	chemical_name	cas_number	hazard_class	default_unit	storage_requirement
Edit Copy Delete	1	Ethanol	64-17-5	Flammable	mL	NULL
Edit Copy Delete	2	Acetone	67-64-1	Flammable	mL	NULL
Edit Copy Delete	3	Sodium Chloride	7647-14-5	Irritant	g	NULL
Edit Copy Delete	4	Hydrochloric Acid	7647-01-0	Corrosive	mL	NULL
Edit Copy Delete	5	Sodium Hydroxide	1310-73-2	Corrosive	g	NULL
Edit Copy Delete	6	Glucose	50-99-7	Low Hazard	g	NULL

Insert Supplier - Adds a new supplier into the system, we can manage new suppliers as new chemical sources available to experiment.

- INSERT INTO Supplier (supplier_id, supplier_name, contact_email)
VALUES (6, 'Northeast Scientific Supply', 'contact@nessupply.com');
DELETE FROM Supplier WHERE supplier_id = 6;

	supplier_id	supplier_name	contact_email	phone
Edit Copy Delete	1	Fisher Scientific	support@fisher.com	800-766-7000
Edit Copy Delete	2	Sigma-Aldrich	support@sial.com	800-325-3010
Edit Copy Delete	3	VWR	support@vwr.com	800-932-5000
Edit Copy Delete	6	Northeast Scientific Supply	contact@nessupply.com	

Hazard Chemicals - Helps users quickly find chemicals with a specific hazard category, this will specifically return all chemicals labeled as flammable.

- SELECT chemical_name, cas_number, hazard_class
FROM Chemical
WHERE hazard_class = 'Flammable';

	chemical_name	cas_number	hazard_class
Edit Copy Delete	Ethanol	64-17-5	Flammable
Edit Copy Delete	Acetone	67-64-1	Flammable

View Batches and its Suppliers - Shows which supplier provided each batch of chemical, Returns batch ID, chemical name, supplier name, and expiration date

- SELECT b.batch_id, c.chemical_name, s.supplier_name, b.expiration_date
  FROM Batch b
  JOIN Chemical c ON b.chemical_id = c.chemical_id
  JOIN Supplier s ON b.supplier_id = s.supplier_id;

batch_id	chemical_name	supplier_name	expiration_date
1	Ethanol	Fisher Scientific	2028-01-10
2	Acetone	Sigma-Aldrich	2028-01-12
3	Sodium Chloride	VWR	2030-01-15
4	Hydrochloric Acid	Fisher Scientific	2027-07-18
5	Sodium Hydroxide	Sigma-Aldrich	2029-01-20
6	Glucose	VWR	2029-01-25

Count Batches by Chemical - Counts how many batches exist for each chemical and Returns each chemical and the number of related batches

- SELECT c.chemical_name, COUNT(b.batch_id) AS total_batches
  FROM Chemical c
  JOIN Batch b ON c.chemical_id = b.chemical_id
  GROUP BY c.chemical_name

chemical_name	total_batches
Acetone	1
Ethanol	1
Glucose	1
Hydrochloric Acid	1
Sodium Chloride	1
Sodium Hydroxide	1

Submission

When you're finished, complete the following steps to submit your work:

- ☐ Export your document with responses as a **PDF file AND save** it **inside** your **documentation** folder. Refer to the following for documentation on how to do this:
  - Google Docs (*File → Download → PDF Document*)
  - Microsoft Word (*File → Save As / Export → PDF*)
  - Pages (*File → Export To → PDF*)
- ☐ Export your updated database as an **SQL file** into the **db** folder. Follow the steps in the Exporting A Database section to export your database into a single SQL file.
- ☐ Upload all your changes to GitHub.
  - ☐ **If you're using GitHub Desktop (GUI)**, complete the Uploading Changes (GitHub Desktop) section to upload your changes from your local device to GitHub.
  - ☐ **If you're using Git (CLI)**, complete the Uploading Changes (GitHub CLI) section to upload your changes from your local device to GitHub.

ONE group member must paste the URL of your GitHub repository in the provided textbox in Brightspace. Click the blue *Submit* button to successfully submit your work for this assignment.

Grading Rubric

You can refer to the **Semester DB Project: Phase II grading rubric** given in Brightspace for this assignment to find details on how your submission will be graded.